

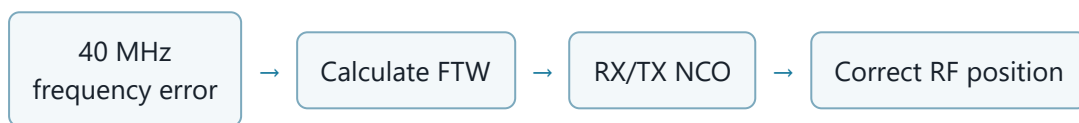
LibreSDR fixed-TCXO NCO guide

Digital correction of the RX and TX tuning frequency

Tezuka/Maia HDL · version 1.0 · 5 September 2026

1. Purpose of the correction

This function is intended only for a LibreSDR fitted with a fixed 40 MHz TCXO, without a usable VCTCXO tuning function. A complex NCO in the FPGA data path compensates the measured oscillator error.



Important: the NCO corrects the RF frequency, but it does not change the physical sample rate. The small sample-rate error in ppm therefore remains.

Safe default state

RX and TX correction are disabled after the bitstream is loaded. Installation and startup are therefore safe; without register writes, the data path remains in bypass mode.

2. Requirements

- The new Libre bitstream is loaded.
- AD936x `xo_correction` remains exactly 40000000.
- A reliable 10 MHz or PPS reference is connected for automatic measurement.
- The commands are run as root and `devmem` is available.

Do not use this fixed-TCXO mode on a board with a working VCTCXO. Such a board should keep the normal PI/DAC control loop active and the digital NCO disabled.

3. Register map

Address	Function	Notes
0x43C00000	DAC control	Bit 0: dac_mode
0x43C00004	Manual/centre DAC code	Low/high 16 bits
0x43C0000C	Reference source	0=10 MHz, 1=PPS_IN, 2=PPS_GPS, 3=none
0x43C00010	Status	Bit 1: ref_present
0x43C00018	Frequency error	Signed: measured – 40,000,000 Hz
0x43C00020	RX FTW	Signed Q0.32
0x43C00024	TX FTW	Signed Q0.32
0x43C00028	NCO control	RX, TX, apply toggle, phase reset

4. Initial setup

Step 1 — freeze the DAC control loop

First set both the manual and centre DAC codes to 0x2800, then enable manual mode. This preserves the existing gain fields.

```
devmem 0x43C00004 32 0x28002800
devmem 0x43C00000 32 0x0001F1F1
```

Step 2 — select and verify the reference

```
# Select one of these values:
devmem 0x43C0000C 32 0x00000000 # external 10 MHz
# devmem 0x43C0000C 32 0x00000001 # PPS_IN
# devmem 0x43C0000C 32 0x00000002 # PPS_GPS

# Wait 2 to 3 seconds, then read status and error:
devmem 0x43C00010 32
devmem 0x43C00018 32
```

Bit 1 must be set in the status word. The error value 0xFFFFF7B represents -133 Hz as a signed 32-bit number: the 40 MHz oscillator is then running 133 Hz slow.

Step 3 — calculate the tuning words

```
actual_xo_hz = 40,000,000 + freq_error
rf_error_hz = requested_rf_hz × freq_error / 40,000,000
actual_fs_hz = requested_fs_hz × actual_xo_hz / 40,000,000

rx_ftw = round( rf_error_hz × 2^32 / actual_fs_hz)
tx_ftw = round(-rf_error_hz × 2^32 / actual_fs_hz)
```

For RX, use the ADC sample rate before Maia/host decimation. For TX, use the final DAC sample rate after any interpolation. If RX and TX use different LO or sample frequencies, calculate their FTWs separately.

Worked example

Given	freq_error=-133, RF=2.5 GHz, Fs=2 MSPS
RF error	-8312.5 Hz
Actual Fs	1,999,993.35 Hz
RX FTW	-17851017 = 0xFEEF9D77
TX FTW	+17851017 = 0x01106289

Step 4 — write the FTWs

```
devmem 0x43C00020 32 0xFEEF9D77 # RX
devmem 0x43C00024 32 0x01106289 # TX
```

Step 5 — test RX first

```
# Enable RX, toggle apply 0→1 and reset phase:
devmem 0x43C00028 32 0x0000000D

# Clear phase reset without applying again:
devmem 0x43C00028 32 0x00000005
```

Check a known carrier. The frequency error should almost disappear. If the error approximately doubles, reverse the sign of the RX FTW; an I/Q swap in the hardware chain can reverse the convention.

Step 6 — enable RX and TX

The apply toggle is 1 after the RX test and must now change to 0:

```
devmem 0x43C00028 32 0x00000003
```

5. Updating and disabling correction

Update the FTWs

Always write both FTWs first. Then invert bit 2 of the control register. With RX and TX enabled, the control value therefore alternates between:

```
0x00000003  # RX+TX enabled, apply toggle=0
0x00000007  # RX+TX enabled, apply toggle=1
```

Keep bit 3 clear during normal drift updates for a phase-continuous change. Recalculate the FTWs after changing the RX LO, TX LO, ADC rate or DAC rate.

Disable the NCO

Enable bits are captured only on an apply transition. First read 0x43C00028, invert bit 2, and write a value with bits 0, 1 and 3 clear:

- Current toggle=1: write 0x00000000.
- Current toggle=0: write 0x00000004.

6. Control-bit definitions

Bit	Name	Meaning
0	RX enable	1 = RX rotation enabled
1	TX enable	1 = TX rotation enabled
2	Apply toggle	Must invert for every new configuration
3	Phase reset	Resets both phases during apply; needed only for initial startup/testing

7. Troubleshooting

Symptom	Check
ref_present remains 0	Check the signal/cable and the value at 0x43C0000C.
freq_error remains zero or stale	Wait for several measurement windows and verify the reference.
The NCO has no effect	Check the enable bits and verify that the apply toggle actually changed.
The error approximately doubles	Reverse the sign of the relevant FTW.
The error returns after reboot	The registers reset at reboot; write the settings again.
RF is correct but sample timing still drifts	Expected: the NCO does not correct the physical sample rate.

Manual devmem settings are suitable for initial setup and testing. For normal operation, a Tezuka software controller should read `freq_error`, obtain the current RX/TX LO and converter sample rates, calculate the FTWs, and update them after tuning, rate or oscillator-drift changes. The FPGA measurement and NCO keep running autonomously between these updates.

Technical register reference: `projects/common/libre-vctcxo-lock/vctcxo_lock.md`